
Bayesian Networks: Independence

CS5491: Artificial Intelligence

ZHICHAO LU

Content Credits: [Prof. Wei's CS4486 Course](#)
and [Prof. Boddeti's AI Course](#)



TODAY

D–Separation



PROBABILITY RECAP

Conditional Probability: $P(x|y) = \frac{P(x,y)}{P(y)}$



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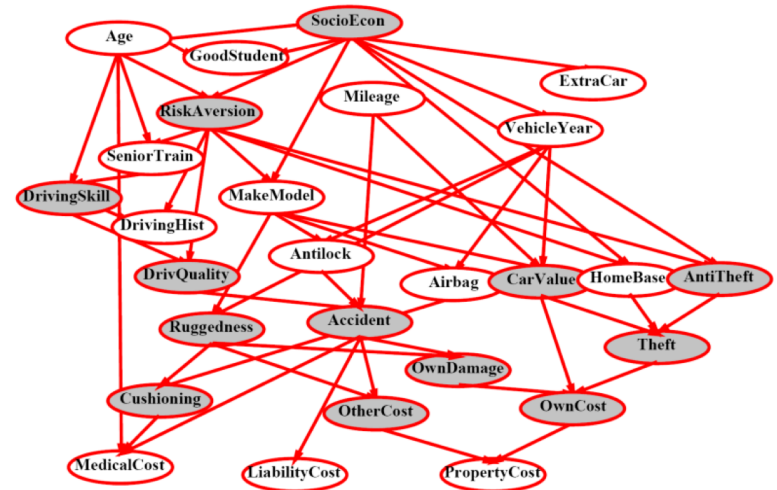
X and Y are conditionally independent given Z if and only if:

$$\forall x, y, z : P(x, y|z) = P(x|z)P(y|z)$$



BAYES NET

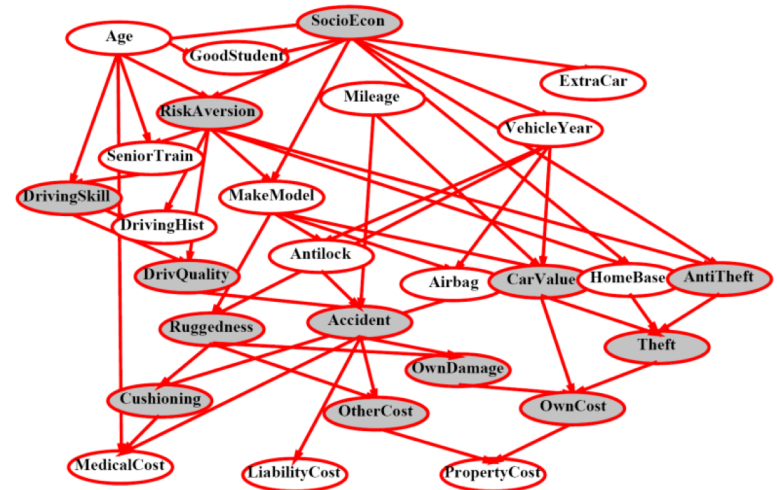
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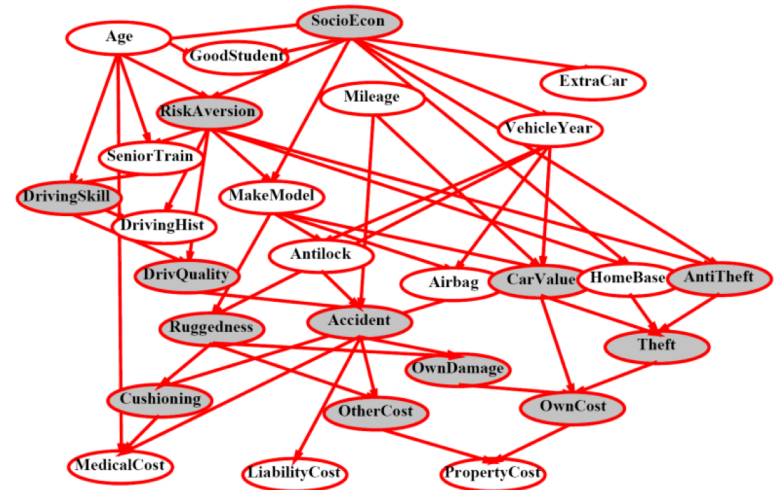


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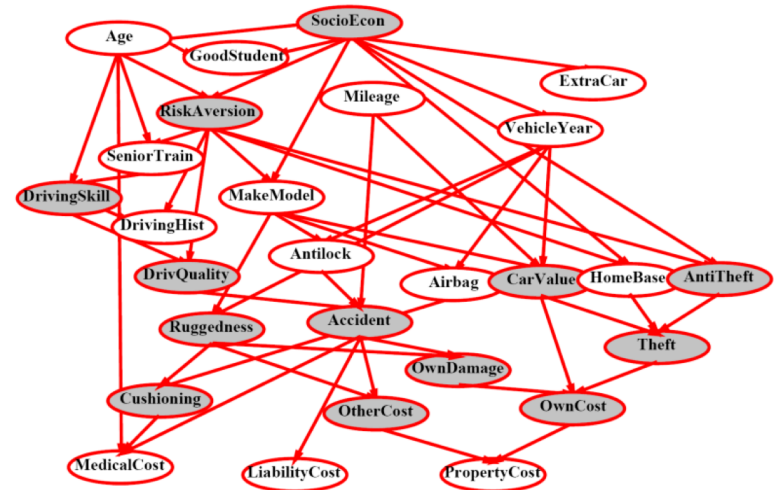


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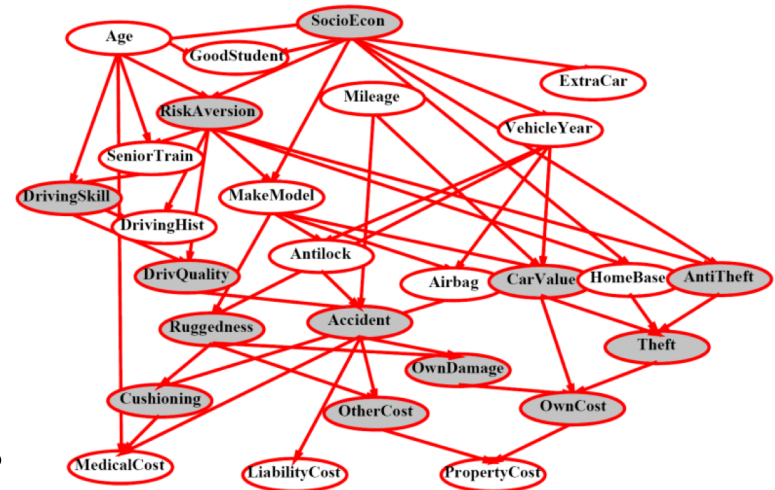


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Questions we can ask:

- Inference: given a fixed BN, what is $P(X|e)$?
- Representation: given a BN graph, what kinds of distributions can it encode?
- Modeling: what BN is most appropriate for a given domain?



BAYES NET SEMANTICS

A directed, acyclic graph, one node per random variable



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A conditional probability table (CPT) for each node



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$$P(X|a_1, \dots, a_n)$$



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Bayes nets implicitly encode joint distributions

- As a product of local conditional distributions
- To see what probability a BN gives to a full assignment, multiply all the relevant conditionals together:

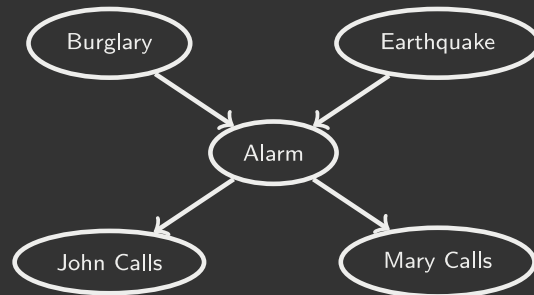
$$P(x_1, \dots, x_n) = \prod_{i=1}^n P(x_i | \text{parents}(x_i))$$



EXAMPLE: ALARM NETWORK

B	P(B)
+b	0.001
-b	0.999

E	P(E)
+e	0.002
-e	0.998



<i>B</i>	<i>E</i>	<i>A</i>	$P(A B, E)$
+b	+e	+a	0.95
+b	+e	-a	0.05
+b	-e	+a	0.94
+b	-e	-a	0.96
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

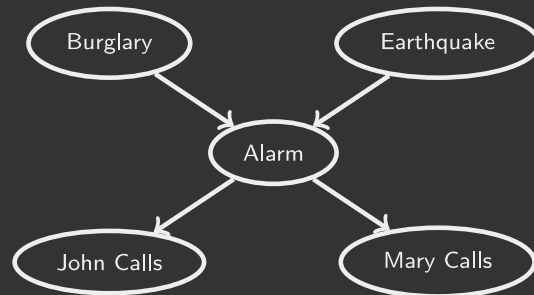
<i>A</i>	<i>J</i>	$P(J A)$
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+a	-j	0.1
-a	+j	0.05
-a	-j	0.95

<i>A</i>	<i>M</i>	$P(M A)$
+a	+m	0.7
+a	-m	0.3
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$$\begin{aligned}
 P(+b, -e, +a, -j, +m) &= P(+b)P(-e)P(+a|+b, -e)P(-j|+a)P(+m|+a) \\
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Both give you the power to calculate

BNs: Huge space savings

Also easier to elicit local CPTs

Also faster to answer queries (coming)



BAYES NET

Representation

Conditional Independences

Probabilistic Inference

Learning Bayes Nets from Data



INDEPENDENCE CONDITIONS



CONDITIONAL INDEPENDENCE

X and Y are independent if:

$$\forall x, y : P(x, y) = P(x)P(y) \Rightarrow X \perp\!\!\!\perp Y$$



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Example: *Alarm* $\perp\!\!\!\perp$ *Fire* | *Smoke*



BAYES NETS: ASSUMPTIONS

Given the graph, assumptions we are required to make in order to define the Bayes net:

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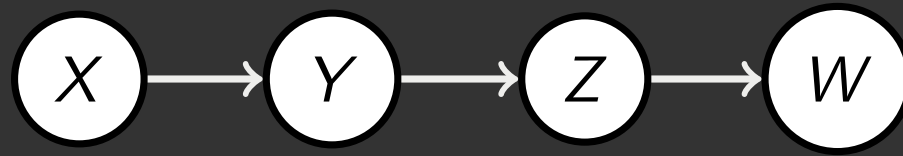
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Important for modeling: understand assumptions made when choosing a Bayes net graph



EXAMPLE



Conditional independence assumptions directly from simplifications in chain rule:

Additional implied conditional independence assumptions?

INDEPENDENCE IN A BN

Important question about a BN:



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 - Addendum: they *could* be independent: how?

D-SEPARATION



D-SEPARATION: OUTLINE

Study independence properties for triples

Analyze complex cases in terms of member triples

D-separation: a condition / algorithm for answering such queries



CAUSAL CHAINS

This configuration is a "causal chain"



low pressure

rain

traffic

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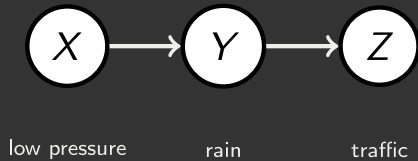
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- Example:
 - Low pressure causes rain causes traffic, high pressure causes no rain causes no traffic
 - In numbers:

$$P(+y|+x) = 1, P(-y|-x) = 1$$

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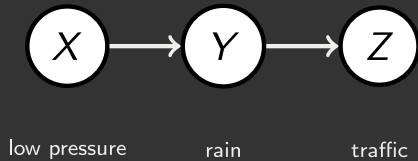
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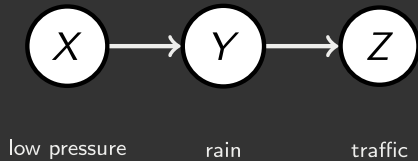
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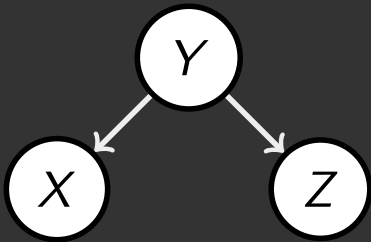
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Evidence along the chain "blocks" the influence

COMMON CAUSE

This configuration is a "common cause"

- Y : project due
- X : forums busy
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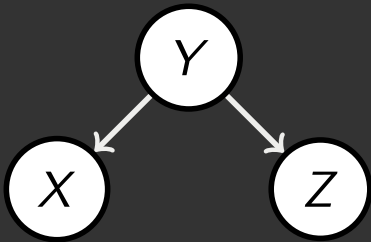
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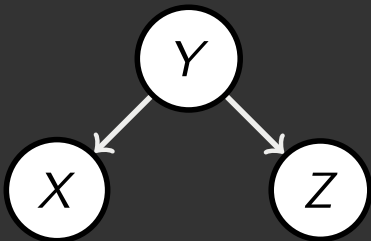
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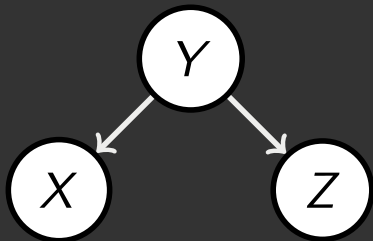


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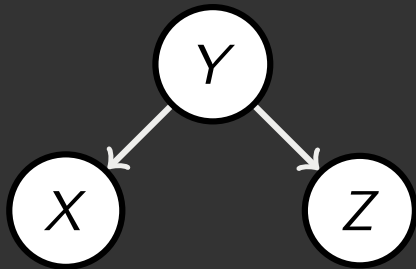
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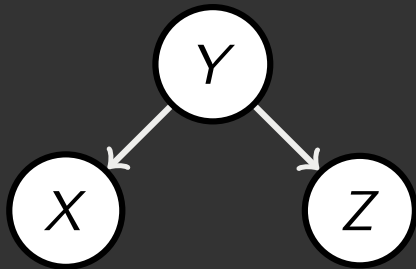
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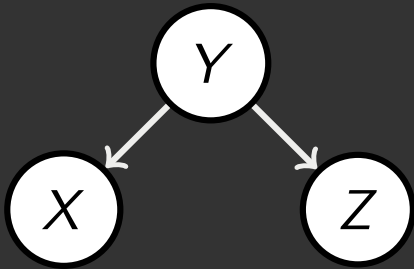


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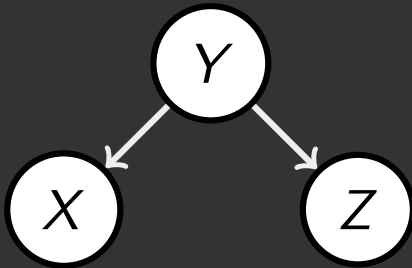
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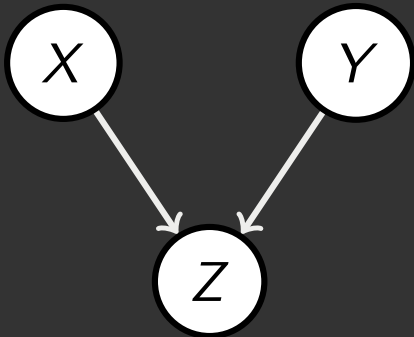
Observing the cause blocks influence between effects

COMMON EFFECT

Last configuration: two causes of one effect (v-structures)

Are X and Y independent?

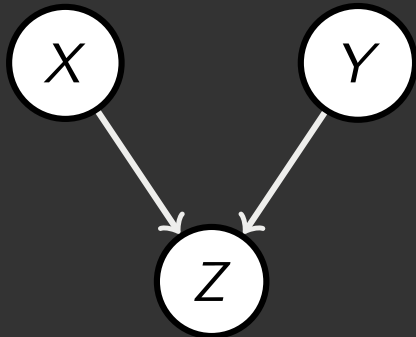
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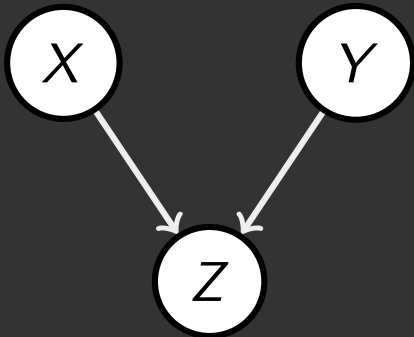
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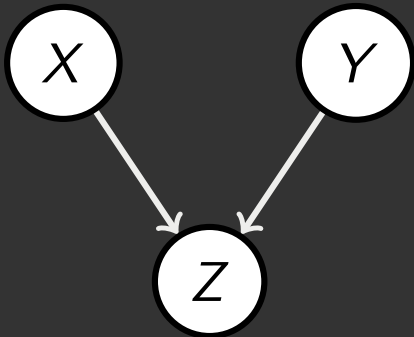
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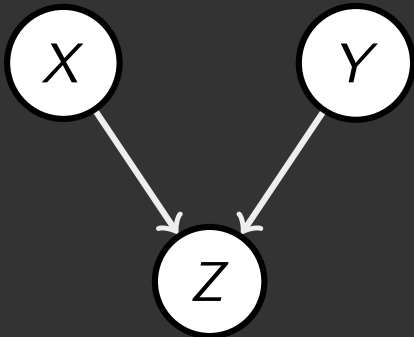
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- Observing an effect **activates** influence between possible causes.

THE GENERAL CASE



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General question: in a given BN, are two variables independent (given evidence)?

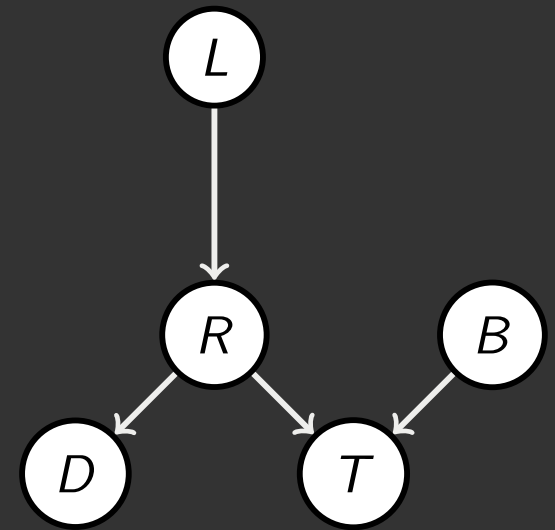
Solution: analyze the graph

Any complex example can be broken into repetitions of the three canonical cases



REACHABILITY

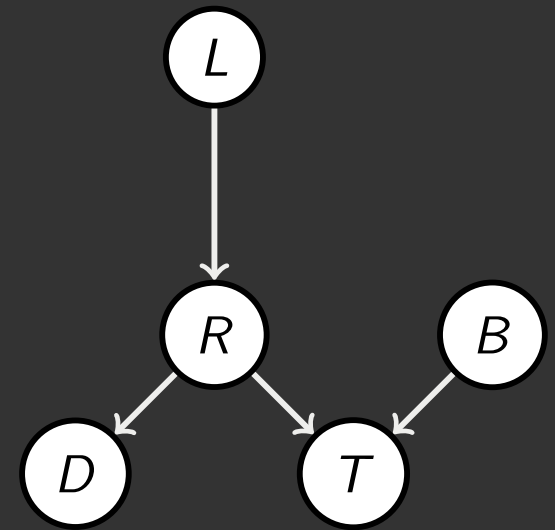
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Attempt 1: if two nodes are connected by an undirected path not blocked by a shaded node, they are reachable.

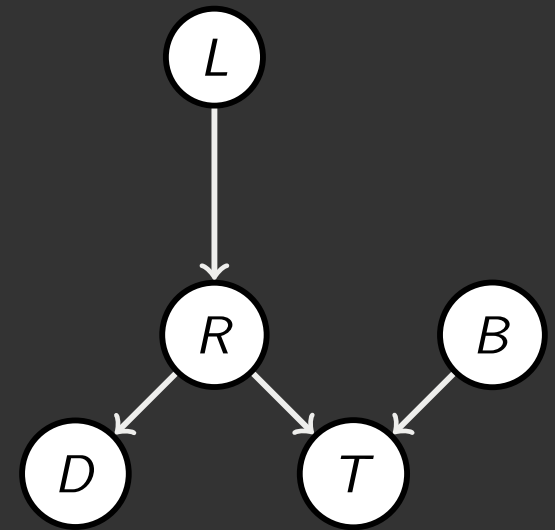


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Almost works, but not quite



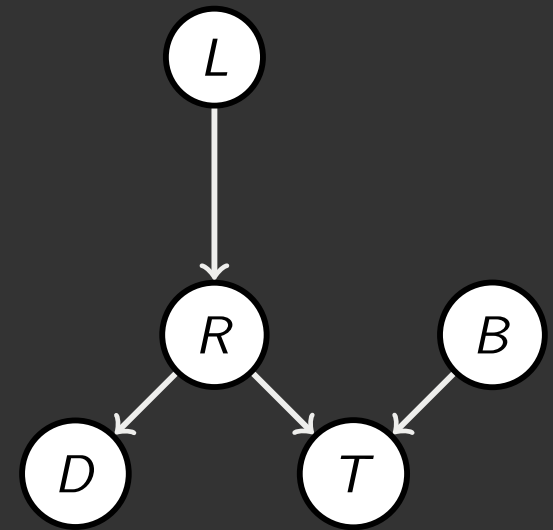
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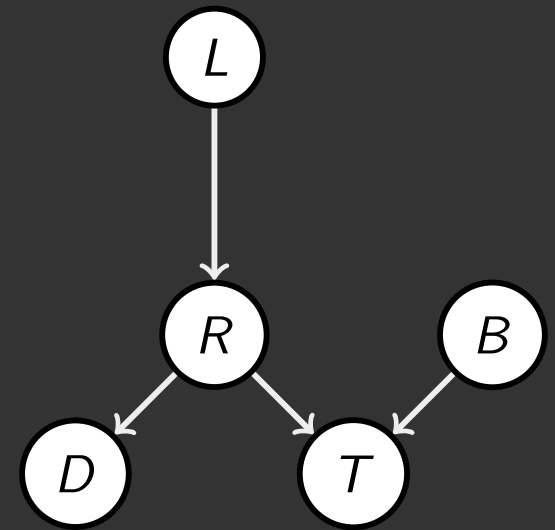
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Almost works, but not quite

- Where does it break?
- Answer: the v-structure at T does not count as a link in a path unless "active"



ACTIVE/INACTIVE PATHS

Question: Are X and Y conditionally independent given evidence variables Z ?

- Yes, if X and Y "d-separated" by Z
- Consider all (undirected) paths from X to Y
- No active paths = independence !!

Active Triples



Inactive Triples



ACTIVE/INACTIVE PATHS

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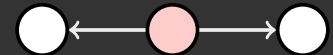
A path is active if each triple is active:

- Causal chain $\mathbf{A} \rightarrow \mathbf{B} \rightarrow \mathbf{C}$ where $\mathbf{B}$ is unobserved (either direction)
- Common cause $\mathbf{A} \leftarrow \mathbf{B} \rightarrow \mathbf{C}$ where $\mathbf{B}$ is unobserved
- Common effect (aka v-structure): $\mathbf{A} \rightarrow \mathbf{B} \leftarrow \mathbf{C}$ where $\mathbf{B}$ or *one of its descendants* is observed

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All it takes to block a path is a single inactive segment

Active Triples



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D-SEPARATION

Query: $X_i \perp\!\!\!\perp X_j \mid \{X_{k_1}, \dots, X_{k_n}\}$



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- Otherwise (i.e. if all paths are inactive), then independence is guaranteed

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STRUCTURE IMPLICATIONS

Given a Bayes net structure, can run d-separation algorithm to build a complete list of conditional independences that are necessarily true of the form

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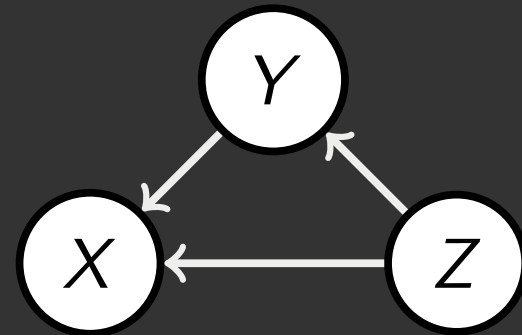
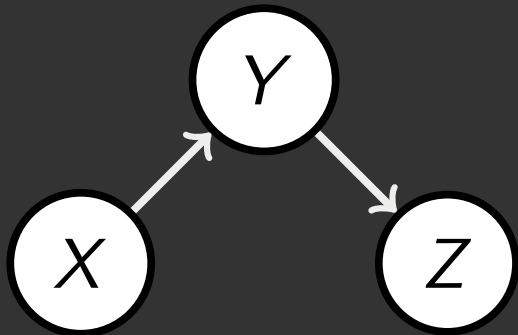
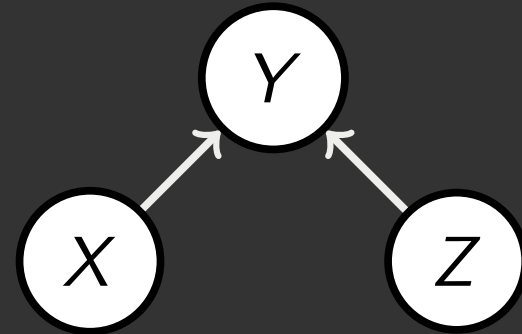
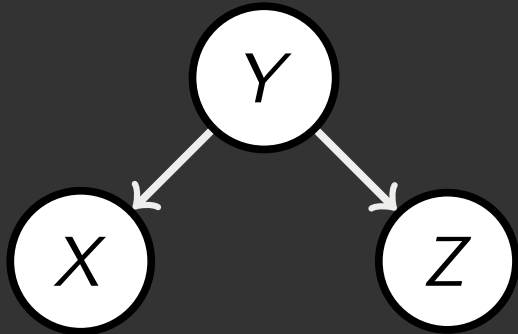
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$$X_i \perp\!\!\!\perp X_j \mid X_{k_1}, \dots, X_{k_n}$$

This list determines the set of probability distributions that can be represented



COMPUTING ALL INDEPENDENCES



TOPOLOGY LIMITS DISTRIBUTIONS

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Full conditioning can encode any distribution



BAYES NETS REPRESENTATION SUMMARY

Bayes nets compactly encode joint distributions

Guaranteed independencies of distributions can be deduced from BN graph structure

D-separation gives precise conditional independence guarantees from graph alone

A Bayes net joint distribution may have further (conditional) independence that is not detectable until you inspect its specific distribution





